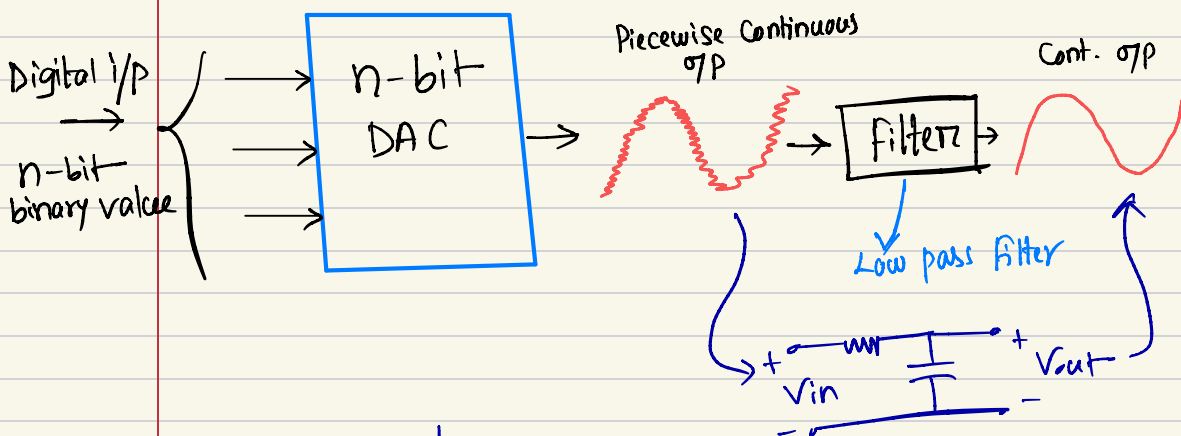


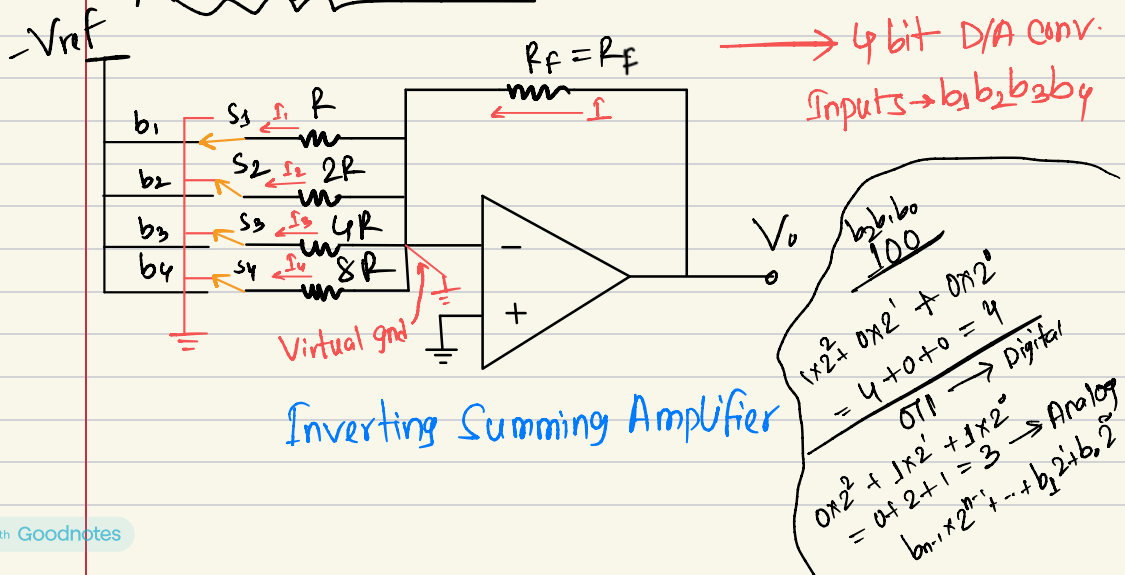
Digital to Analog Conversion



D/A Converters

- Binary Weighted Resistor D/A converter. ✓
- R-2R Ladder ✗

Binary weighted D/A converter:



- If b_1 is 1 → S_1 is connected to the resistor.
 → If b_1 is 0 → S_1 is connected to ground.

here, $I = \frac{V_o - 0}{R_F} = \frac{V_o}{R_F}$

if $b_1 = 1 \rightarrow I_1 = \frac{0 - (-V_{ref})}{R} = \frac{V_{ref}}{R}$
 if $b_1 = 0 \rightarrow I_1 = \frac{0 - 0}{R} = 0$
 $\therefore I_1 = b_1 \frac{V_{ref}}{R}$

Similarly $\rightarrow I_2 = b_2 \frac{V_{ref}}{2R}$

$I_3 = b_3 \frac{V_{ref}}{4R}$

$I_4 = b_4 \frac{V_{ref}}{8R}$

We can write, $I = I_1 + I_2 + I_3 + I_4$

$\Rightarrow \frac{V_o}{R_F} = b_1 \frac{V_{ref}}{R} + b_2 \frac{V_{ref}}{2R} + b_3 \frac{V_{ref}}{4R} + b_4 \frac{V_{ref}}{8R}$

$\Rightarrow V_o = V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \times \frac{R_F}{R}$

Similarly

→ for 3 bits $\rightarrow V_o = V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right) \times \frac{R_F}{R}$

→ for 5 bits $\rightarrow V_o = V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} + \frac{b_5}{16} \right) \times \frac{R_F}{R}$

Practice Problem

Let 4 bit Binary weighted Resistor have
 $V_{ref} = 16V$, $R_F = R$

i) if i/p is 1010, $V_o = ?$

ii) What is the maximum value of o/p?

Soln:

i) $b_1 b_2 b_3 b_4$ $V_{ref} = 16V$, $R_F = R$
1010

$$\begin{aligned}\therefore V_o &= V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \times \frac{R_F}{R} \\ &= 16 \left(1 + 0 + \frac{1}{4} + 0 \right) \\ &= 20V \quad (\text{Ans.})\end{aligned}$$

ii) max if $b_1 b_2 b_3 b_4 = 1111$

$$\begin{aligned}\therefore V_o &= 16 \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) \times \frac{R_F}{R} \\ V_{max} &= 30V \quad (\text{Ans.})\end{aligned}$$

PP-2

Design 6 bit BWR D/A where $V_{max} = 10V$
Draw the ckt also.

let 4 bit

* Step size $\rightarrow$ 1 LSB value $\rightarrow V_o$ corresponds to 0001

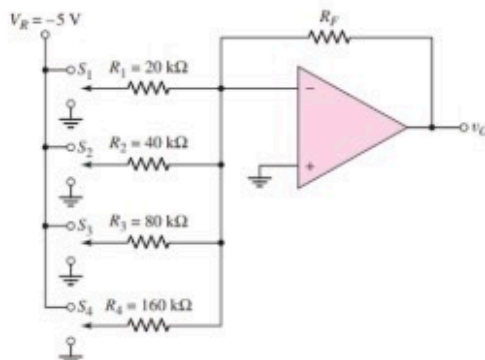
* 1 MSB value $\rightarrow V_o$ corresponds to 1000

* Max o/p voltage $\rightarrow V_o$ " " 1111 $\rightarrow V_{max}$

* Min o/p " $\rightarrow V_o$ " " 0000 $\rightarrow V_{min}$

PP-3

- (a) Determine the output voltage, v_o in V of the 4-bit weighted-resistor D/A in the following figure for input = 1010 and input = 1100? Assume $R_f = 5k\Omega$.
- (b) Identify the maximum allowed tolerance ($\pm$ percent) in the value of R_1 so that the maximum error in the output is limited to $\pm \frac{1}{2}$ LSB quantized voltage value? (hint: Do two separate calculations for output error being $+\frac{1}{2}$ LSB and $-\frac{1}{2}$ LSB. Then, calculate the percentage change in R_1 in these two cases.)



a) Input = 1010; $R_f = 5k\Omega$

$$v_o = 5 \times \frac{5}{20} + 5 \times \frac{5}{80} = \boxed{1.5625V}$$

Input = 1100; $R_f = 5k\Omega$

$$v_o = 5 \times \frac{5}{20} + 5 \times \frac{5}{40} = \boxed{1.875V}$$

b) $1 \text{ LSB} = 5 \times \frac{5}{160} = 0.15625V$

$$\Rightarrow \frac{1}{2} \text{ LSB} = 0.078125V$$

$$1 \text{ MSB} = 1.25V \rightarrow$$

$$v_o = 1.25 \pm 0.078125V$$

$$\pm 1.328125V \text{ or } 1.171875V$$

$$-5 \times \frac{5}{R_1} = 1.328125V$$

or

$$-5 \times \frac{5}{R_1} = 1.171875V$$

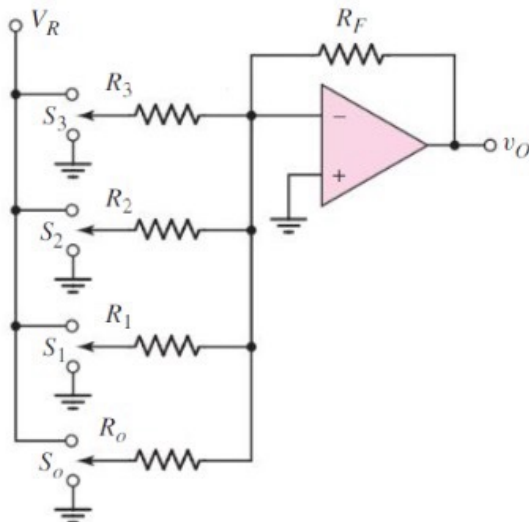
$\Rightarrow$

$$R_1 = 18.823k\Omega \rightarrow \% \text{ change in } R_1 = \frac{20 - 18.823}{20} \times 100 = \boxed{5.88\%}$$

$$\text{or } 21.33k\Omega \rightarrow \% \text{ change in } R_1 = \frac{21.33 - 20}{20} \times 100 = 6.67\%$$

$$\boxed{\text{Max tolerance} \rightarrow 5.88\%}$$

8. For the binary weighted DAC shown below, answer the following questions:



Here, $V_R = -8V$, $R_3 = R_f$ & Input bit sequence: $S_3 S_2 S_1 S_0$

(a)	CO2	Show the output voltages for all possible input cases 'XXX0' in a table, where X may be 0 or 1.	[10]
(b)	CO2	What is the ratio of maximum output voltage & LSB(for input 0001) voltage?	[4]
(c)	CO2	What should be the ratio $\frac{R_3}{R_f}$, if we want the output voltage for input '1010' to be 5V?	[6]

Solution:

a)

Input	Output = $ V_R \times (B_3 \times 1 + B_2 \times 0.5 + B_1 \times 0.25)$
0000	0
0010	2
0100	4
0110	6
1000	8
1010	10
1100	12
1110	14

b) Maximum voltage = $|V_{ref}|(1 + 0.5 + 0.25 + 0.125) = 15V$

LSB voltage = $|V_{ref}|(0.125) = 1V$

So, ratio = 15:1

c) Output voltage of '1010' = $|V_{ref}|(\frac{R_f}{R_3} + 0.25 \times \frac{R_f}{R_3}) = 5$

Thus, $\frac{R_f}{R_3} = 0.5$